21-242 Recitation 12

Subspaces — Q2. Kernel and image are subspaces

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

Show that $\ker(T)$ and $\mathrm{im}(T)$ are subspaces if T is a linear transformation.

Course-level solution

Let $T : V \to W$ be linear.
Kernel. $T(\vec{0}) = \vec{0}$, so $\vec{0} \in \ker T$. If $\vec{v}, \vec{w} \in \ker T$ and $a, b \in \mathbb{F}$, then

$$T(a\vec{v} + b\vec{w}) = aT(\vec{v}) + bT(\vec{w}) = \vec{0},$$

so $a\vec{v} + b\vec{w} \in \ker T$.
Image. $\vec{0} = T(\vec{0}) \in \mathrm{im}\, T$. If $\vec{w}_i = T(\vec{v}_i)$ and $a, b \in \mathbb{F}$, then $a\vec{w}_1 + b\vec{w}_2 = T(a\vec{v}_1 + b\vec{v}_2) \in \mathrm{im}\, T$.